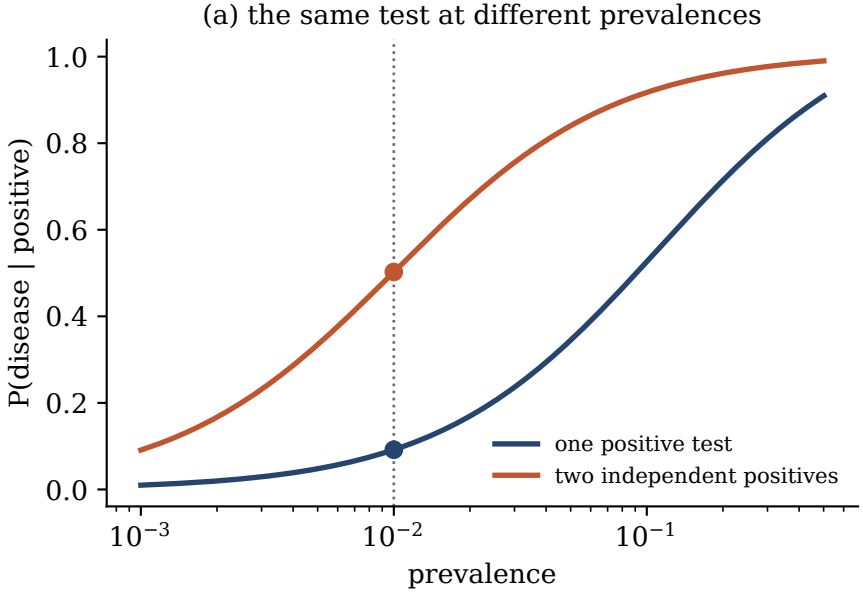
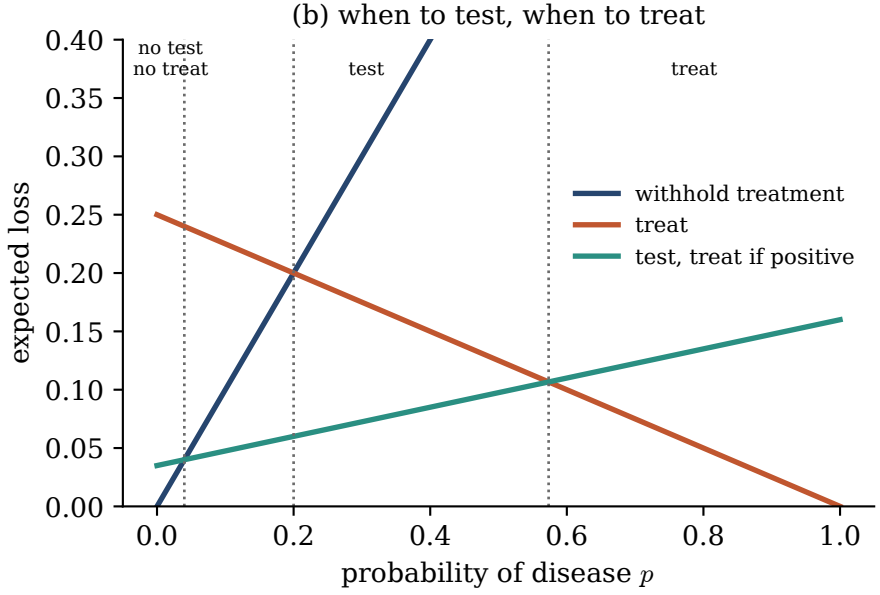

(a) the same test at different prevalences
P(disease | positive)
1.0
0.8
0.6
0.4
0.2
0.0
$10^{-3}$
$10^{-2}$
$10^{-1}$
prevalence
one positive test
two independent positives


(b) when to test, when to treat
expected loss
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
no test
no treat
test
treat
withhold treatment
treat
test, treat if positive
0.0
0.2
0.4
0.6
0.8
1.0
probability of disease $p$